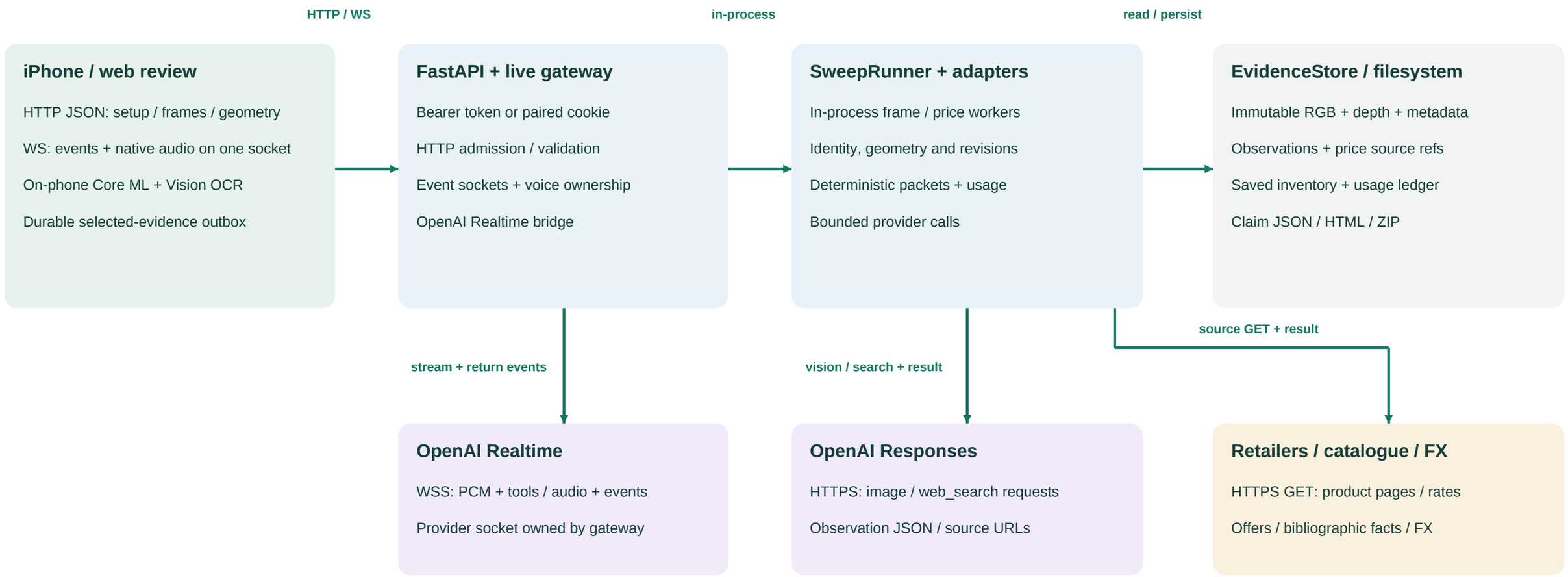
Request / response architecture

One native client, one FastAPI process, external model and source APIs. Internal modules are not separate deployed microservices.



Routes show the caller. Reverse responses/events use the same connection; the numbered sequences below show each direction separately.

Detailed sequence index

F01 Open a session and connect voice

F02 Spoken setup, readiness and capture handoff

F03 Stream audio, acknowledge playback and interrupt

F04 Select, retain and upload camera evidence

F05 Recognize a frame and publish inventory

F06 Measure objects and update RoomPlan geometry

F07 Search, verify and apply a price

F08 Read inventory and apply corrections

F09 Finish, generate and download claim evidence

F10 Compare camera-only with LiDAR

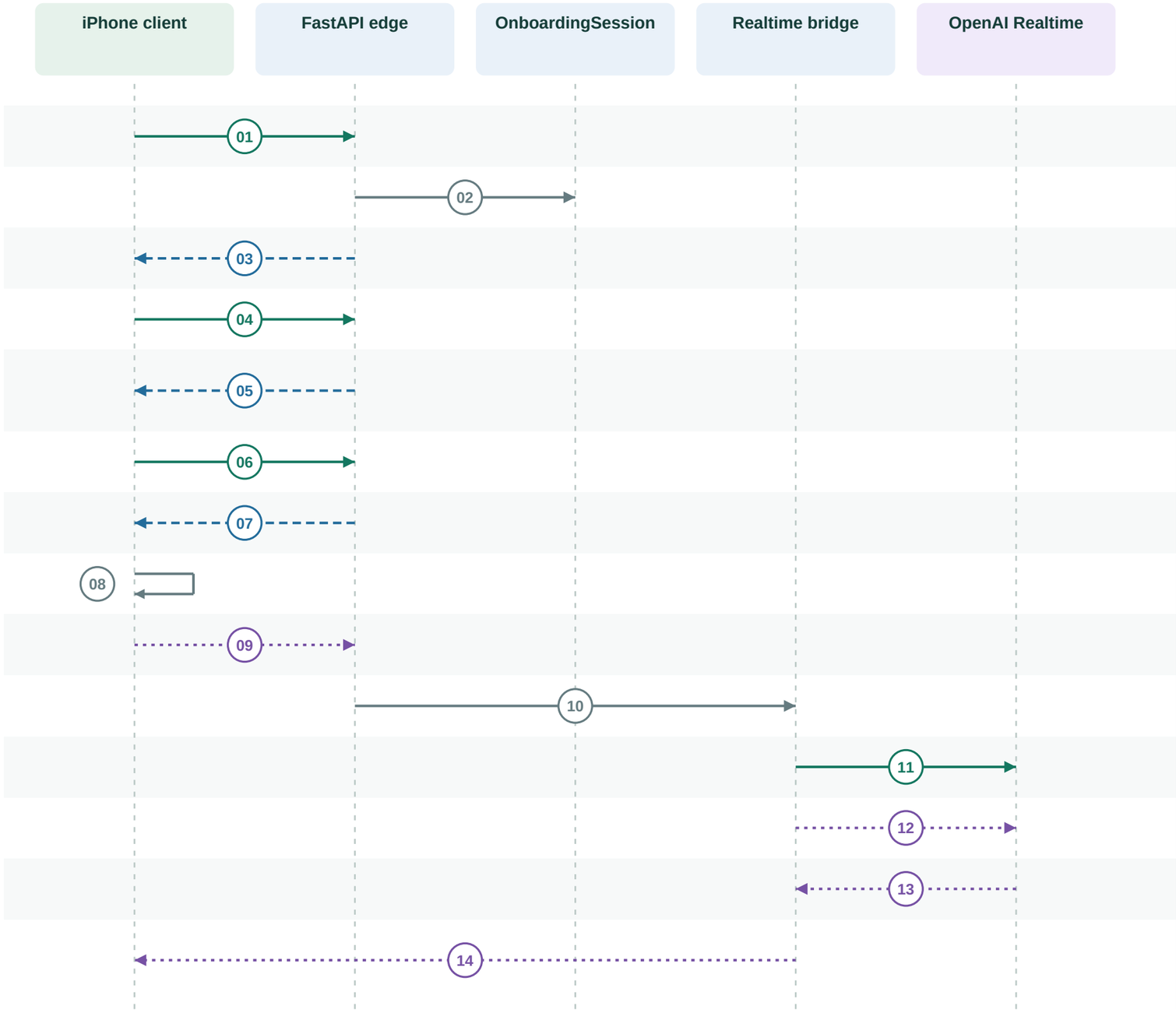
F11 Reprice the same books in a second market

F12 Reconnect, recover and account for usage

Green solid: network request | Blue dashed: network/result response | Purple dotted: asynchronous event | Grey solid: internal call

F01 Open a session and connect voice

HTTP creates setup state; the client WebSocket and provider WebSocket are separate connections.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 POST /api/onboarding
JSON: device, capture_supported (LiDAR), camera_supported.

02 Create bounded in-memory setup session
Expire/consume old sessions; at most 8 live setup sessions; TTL 600 s.

03 200 {id, expires_in_s: 600}
Alternative: 409 conversation limit; common auth/schema errors apply.

04 GET /api/health
Read advertised OpenAI voice transport/readiness; no provider request.

05 200 capability/configuration JSON
Require voice_transport=realtime, assistant_provider=openai, realtime_ready=true.

06 WS /api/onboarding/{id}/live
Authenticated WebSocket handshake; startVoice:false in native client.

07 WebSocket accepted; listener attached
Independent event sender; stored setup events may be replayed.

08 Prepare AVAudioEngine and event tasks
Settle voice-processing audio graph before asking for paid voice.

09 start_voice
JSON control event; await voice_ready or voice_error, not an HTTP reply.

10 Select OpenAILiveBridge and claim voice owner
A second voice owner gets voice_error; event socket can remain open.

11 WSS api.openai.com/v1/realtime?model=...
Server credential; configured gpt-realtime-2.1 in submission configuration.

12 session.update
24 kHz PCM, Marin, semantic VAD, tools, asynchronous input transcription.

13 session.updated
Provider acknowledges configuration; wait bounded to 20 s.

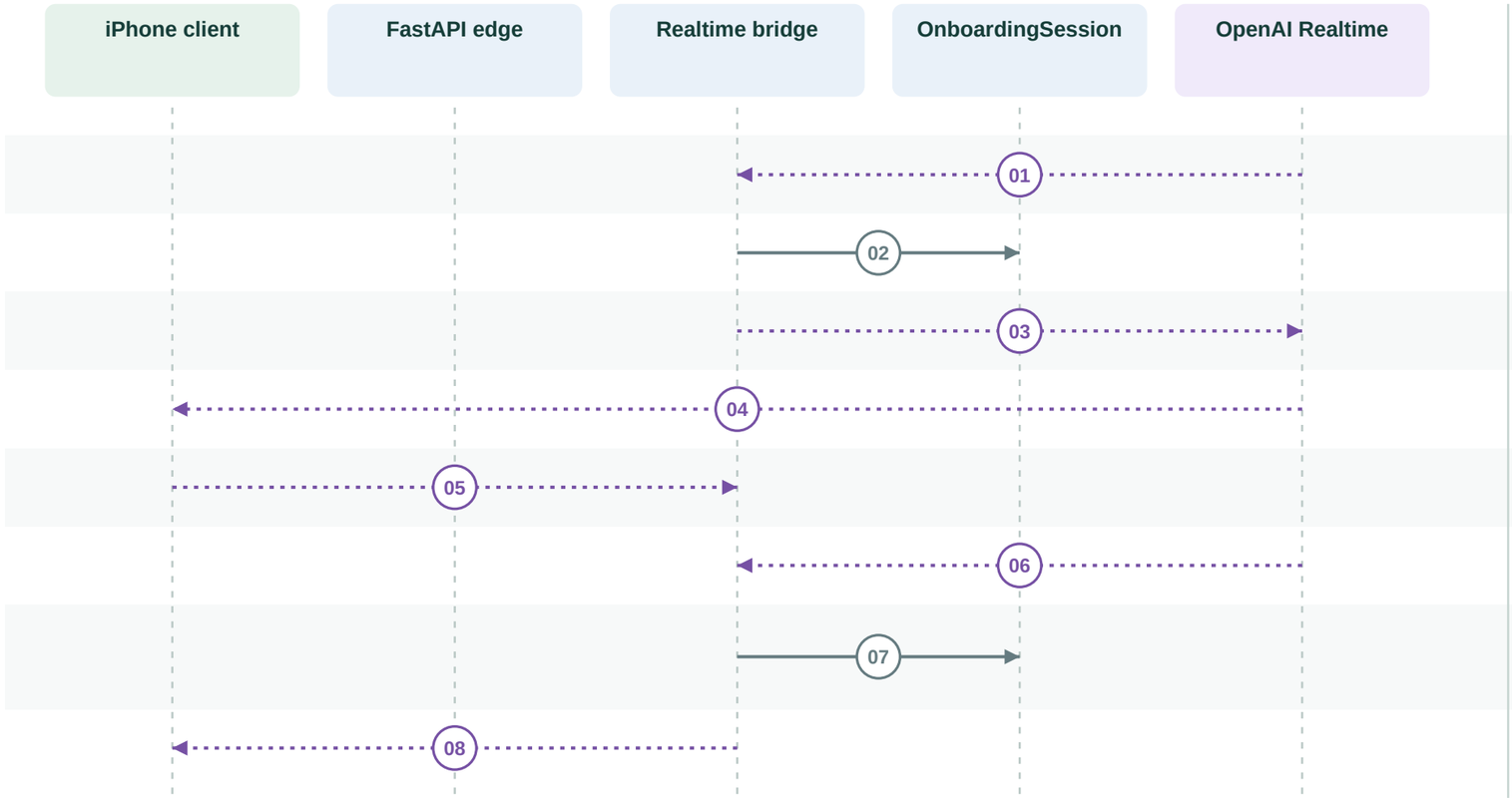
14 voice_ready (via FastAPI event sender)
Server enqueues voice_ready and requests the greeting; native enables microphone when voice_ready arrives.

BOUNDARY The native app sends a bearer pairing token; it does not call /api/pair. Health reports configuration, not funded access. A voice_ready event, not socket open, enables microphone sending.

SOURCE claim_agent/server.py:262 | claim_agent/server.py:422 | claim_agent/openai_live.py:633 | ios/LibraryClaim/CaptureController.swift:584

F02 Spoken setup, readiness and capture handoff - part 1/2

Tool requests are provider events; the server validates them and changes setup state in process.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 **response.function_call_arguments.done**
present_setup: language, country, currency, capture_mode, prompt templates.

02 **Validate presented preferences and owner turn**
Render readback server-side; select LiDAR when supported, otherwise camera.

03 **conversation.item.create + response.create**
Function output by call_id; spoken full readback requested with tools disabled.

04 **Assistant audio via bridge and API**
response.output_audio.delta -> audio; transcript/done tracked separately.

05 **playback via API: actual played_ms**
Audio completion + output transcript + played coverage open confirmation gate.

06 **Later owner turn: configure_session**
Matching preferences with confirmed:true after delivered readback.

07 **Store confirmed preferences in memory**
Onboarding session only; emit setup_confirmed and schedule separate readiness question.

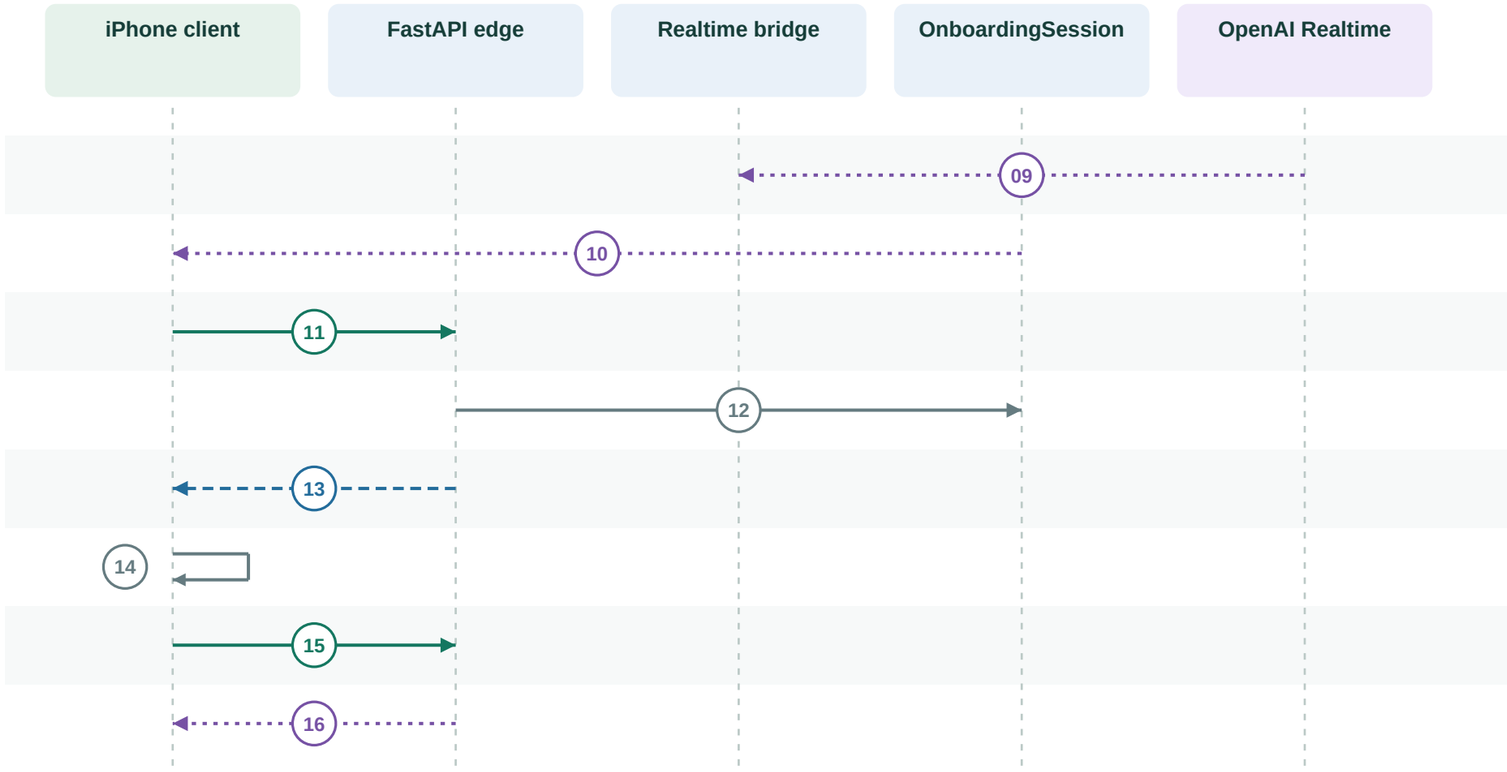
08 **setup_confirmed {preferences}; readiness audio**
Phone acknowledges actual readiness-question playback before next gate.

BOUNDARY Confirmation intent is model-interpreted. Code enforces later intelligible owner turns, gate versions and matching arguments; playback must cover readback and readiness. Preview may run, evidence capture does not.

SOURCE claim_agent/openai_live.py:187 | claim_agent/openai_live.py:426 | claim_agent/onboarding.py:110 | claim_agent/server.py:211 | ios/LibraryClaim/CaptureController.swift:205

F02 Spoken setup, readiness and capture handoff - part 2/2

Tool requests are provider events; the server validates them and changes setup state in process.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

09 **Later owner turn: request_capture {ready:true}**
Validate readiness phase, intelligible turn and physical sensor capability.

10 **capture_requested via API**
Native closes onboarding voice/events before creating the sweep.

11 **POST /api/sweeps**
JSON: device, onboarding_id. Retry lookup first checks persisted handoff.

12 **Consume setup and copy preferences / usage**
Persist onboarding_id with new sweep; close setup voice, refresh copied usage.

13 **200 ClaimPacket snapshot**
Same onboarding ID recovers existing sweep; invalid/expired handoff is 409.

14 **Start shared AR/RoomPlan session**
Create capture outbox and clock; choose mode from returned sweep state.

15 **WS /api/sweeps/{id}/live**
New sweep event connection; a new start_voice follows F01 provider setup.

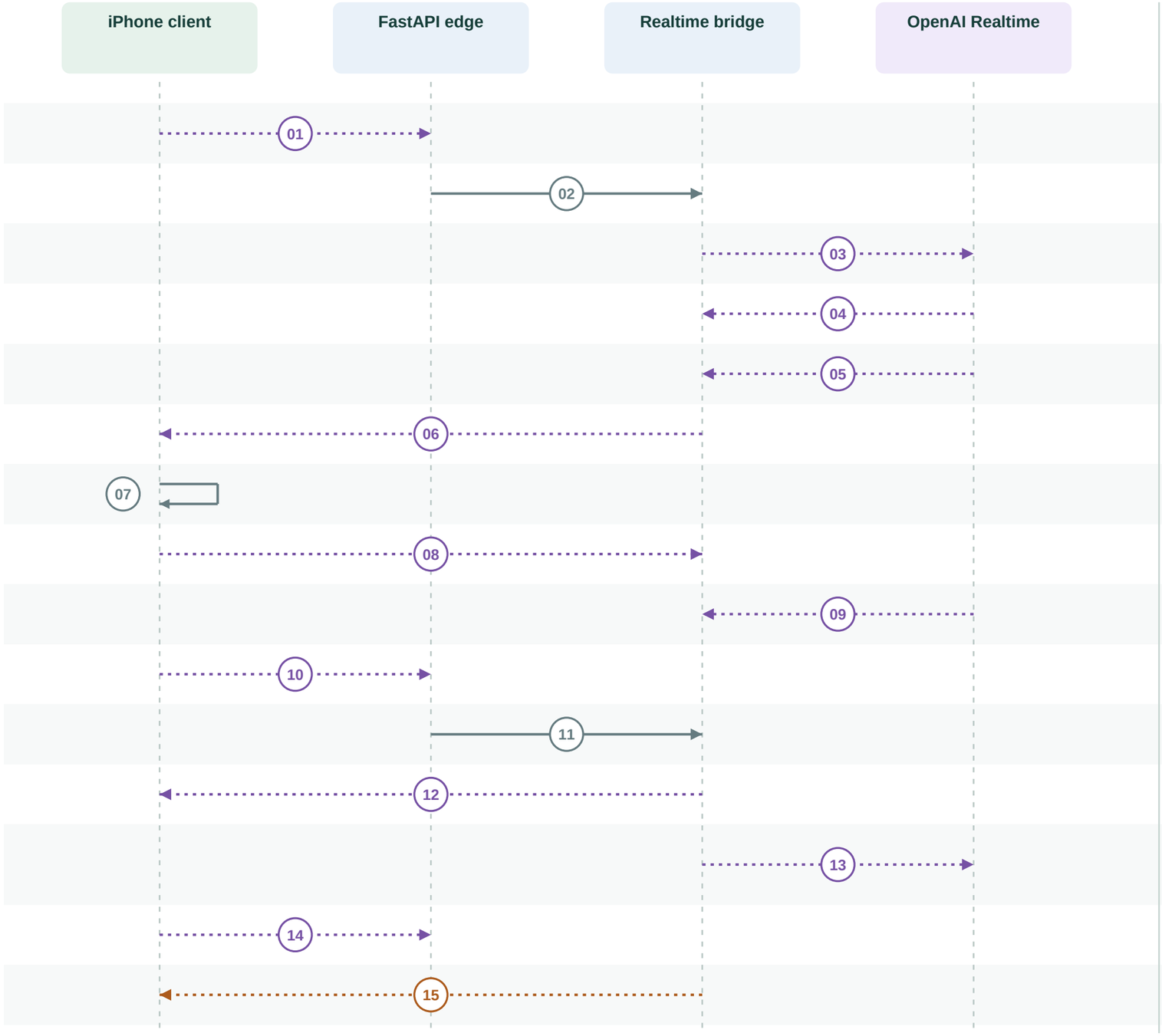
16 **inventory {packet}; later voice_ready**
Initial snapshot is independent of new provider voice readiness.

BOUNDARY Confirmation intent is model-interpreted. Code enforces later intelligible owner turns, gate versions and matching arguments; playback must cover readback and readiness. Preview may run, evidence capture does not.

SOURCE claim_agent/openai_live.py:187 | claim_agent/openai_live.py:426 | claim_agent/onboarding.py:110 | claim_agent/server.py:211 | ios/LibraryClaim/CaptureController.swift:205

F03 Stream audio, acknowledge playback and interrupt

Audio and transcript events stream asynchronously; generation completion does not prove playback.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 audio {pcm_base64}
Signed PCM16 mono at 16 kHz; microphone enabled only after voice_ready.

02 Validate and enqueue audio
Even byte count; at most 32,000 bytes per native audio event.

03 input_audio_buffer.append {audio}
Stateful resampling to 24 kHz; provider VAD controls ordinary turn creation.

04 input_audio_buffer.speech_started / speech_stopped / committed
Independent transcription.completed and response events follow.

05 response.output_audio.delta
PCM16 mono 24 kHz plus response/item/content identifiers.

06 audio via API event socket
pcm_base64, sample_rate=24000, item_id, content_index, audio_end_ms.

07 Schedule playback on AVAudioPlayerNode
Track cumulative played duration with dataPlayedBack callbacks.

08 playback via API
item_id, content_index, played_ms: receipt of audio actually played.

09 Transcript and response.done events
Forward transcript / turn_complete separately; record reported usage.

10 ALT manual: interrupt_voice
Phone stops queued playback and temporarily suppresses late audio.

11 Handle interruption request
Spoken barge-in instead starts from provider input_audio_buffer.speech_started.

12 interrupted via API
Publish interrupted first; invalidate pending work and clear stale playback state.

13 response.cancel; conversation.item.truncate
Manual: response.cancel only if active; truncate only when an unplayed audio tail exists.

14 Optional stop_voice
Close provider bridge while retaining event-only socket where possible.

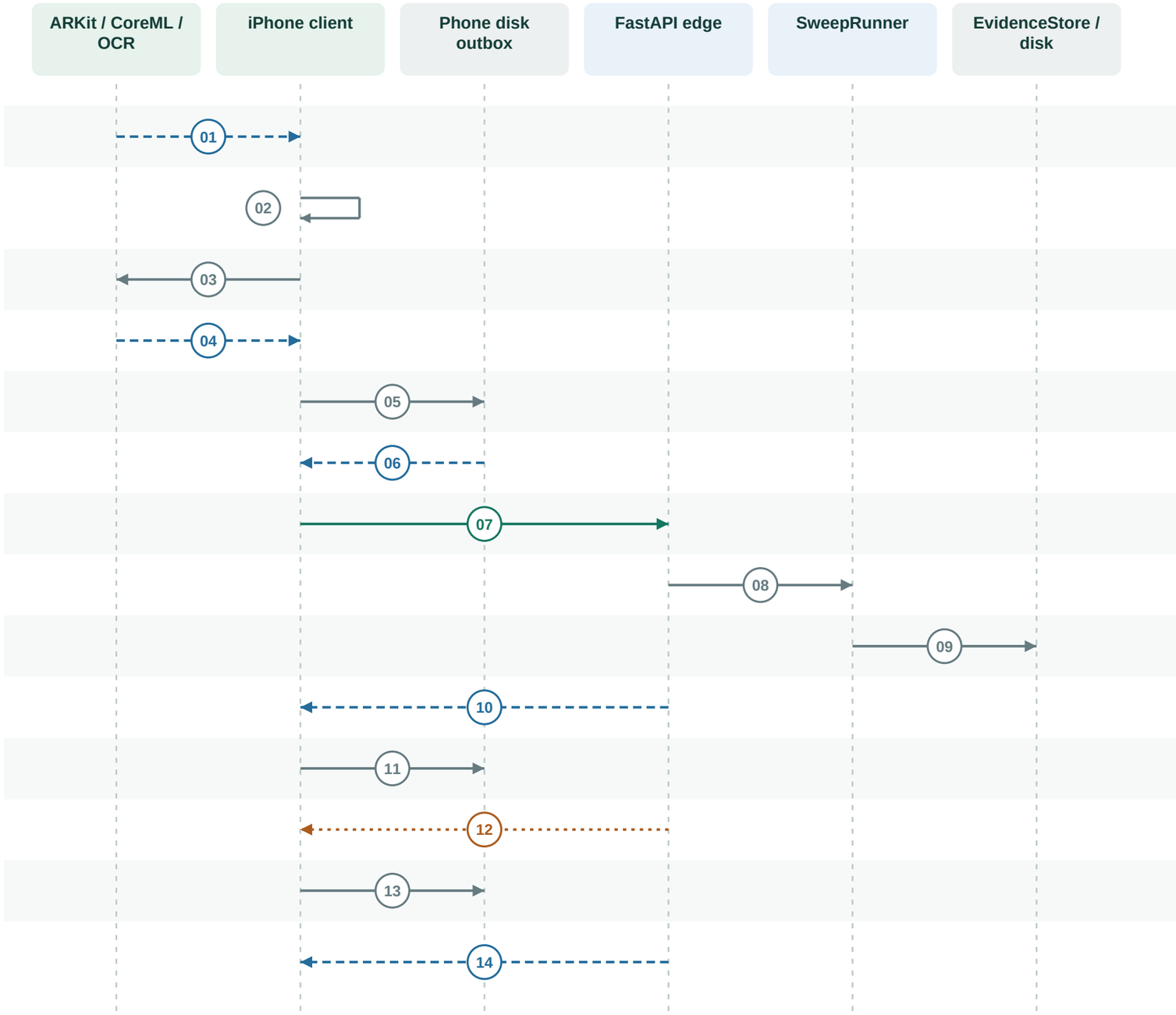
15 ALT unrecovered provider failure: voice_error
Sanitized reconnect/billing guidance; preserve usage and review reason.

BOUNDARY The active protocol has no sequential STT -> text model -> TTS chain. Input transcription assists action gates asynchronously. No routine uploaded JPEG is forwarded to Realtime. Disconnect does not erase saved capture evidence. Native audio and inventory share one client WebSocket; the gateway owns a separate provider WebSocket.

SOURCE ios/LibraryClaim/LiveAudio.swift:53 | ios/LibraryClaim/LiveAudio.swift:135 | claim_agent/server.py:448 | claim_agent/openai_live.py:286 | claim_agent/openai_live.py:495

F04 Select, retain and upload camera evidence

An HTTP receipt acknowledges immutable admission or suppression. Recognition completes later (F05).



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 Local detector proposals + quality + fingerprint
Core ML full frame/detail crops; display provisional dotted outlines.

02 Apply selection and concurrency gates
Tracking, blur/exposure, novelty, pending count, time interval, depth improvement.

03 Encode selected frame and run Vision OCR
Raw-axis JPEG + calibration/pose + OCR; LiDAR depth/confidence if supported.

04 Encoded FrameUpload payload
local_analysis=cloud_requested/unavailable; no Apple language model call.

05 Atomically save pending/{frame_id}.json
Immutable bytes before network; 1 GB outbox budget.

06 Earliest pending frame payload
One uploader sends saved frames sequentially; retries reuse exact bytes.

07 POST /api/sweeps/{id}/frames
metadata, jpeg_base64, depth_base64, confidence_base64; bearer auth.

08 Validate payload and compute receipt signature
Dimensions, image/depth lengths, same AR session and capture state.

09 Persist evidence, receipt and queued state
New frames enter bounded queue; JPEG content is hash-addressed.

10 200 {accepted:true, processing:"queued", ...}
duplicate:false and queue_depth for new frames; not a recognition result.

11 Move pending file to uploaded/
Preserve bytes; accepted frame unlocks live geometry upload.

12 ALT full queue: 429 + Retry-After: 3
detail.processing=retry_backpressure, accepted:false, retry_after_s:3.

13 ALT retain pending; delay and retry same bytes
Other upload errors also retain pending evidence; no second frame ID.

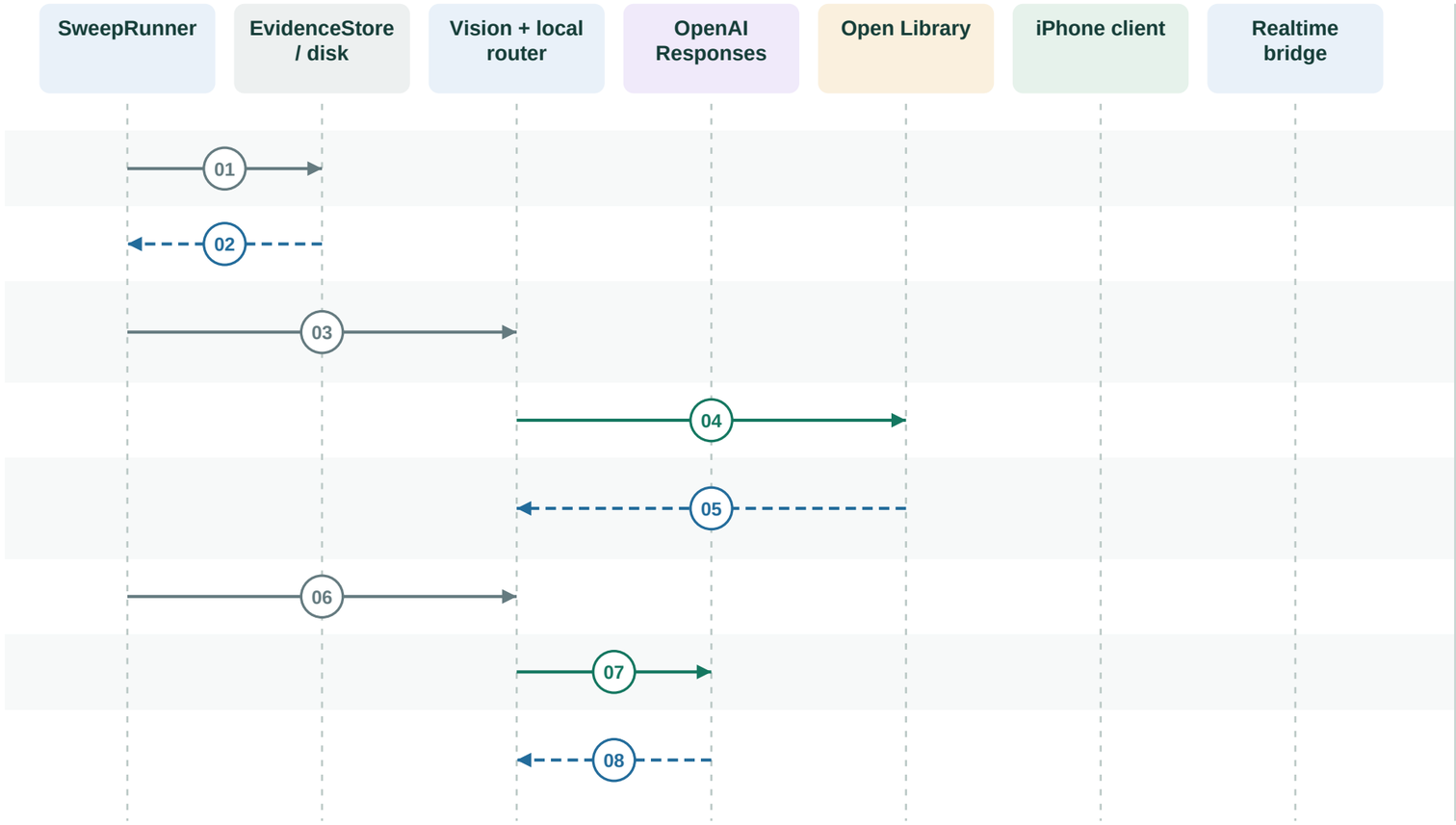
14 ALT duplicate: 200 with duplicate:true
processing=duplicate_suppressed or queued_geometry; original receipt may replay.

BOUNDARY The phone retains acknowledged bytes. Same-ID retries cannot change metadata/calibration. Similar-image suppression stores metadata, not a second JPEG; exact RGB plus new depth may queue geometry-only refinement.

SOURCE ios/LibraryClaim/LocalRecognitionModels.swift:142 | ios/LibraryClaim/FrameEncoder.swift:48 | ios/LibraryClaim/EvidenceOutbox.swift:23 | claim_agent/pipeline.py:154 | claim_agent/server.py:361

F05 Recognize a frame and publish inventory - part 1/2

Frame workers run concurrently; publication order can differ from capture order.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 Dequeue frame; read hash-verified JPEG
Duplicate-geometry job instead reuses source polygons with its own depth.

02 Saved image + calibrated evidence
Receipt integrity establishes bytes, not recognition correctness.

03 Resolve local OCR / proposal evidence
Apply independent text and detector support; optional catalogue lookup precedes cloud reservation.

04 OPTIONAL GET openlibrary.org/search.json
Compatible book clues only: title, author, limit=3, selected metadata fields.

05 OPTIONAL candidate bibliographic metadata
Corroboration only; catalogue ISBN/edition does not become observed evidence.

06 Quality / novelty / budget / cooldown decision
Reserve and persist any cloud attempt before a billable provider request.

07 POST /v1/responses (standard OpenAI base)
gpt-6-astra; upright selected JPEG + hints; store:false; strict ProposedScan.

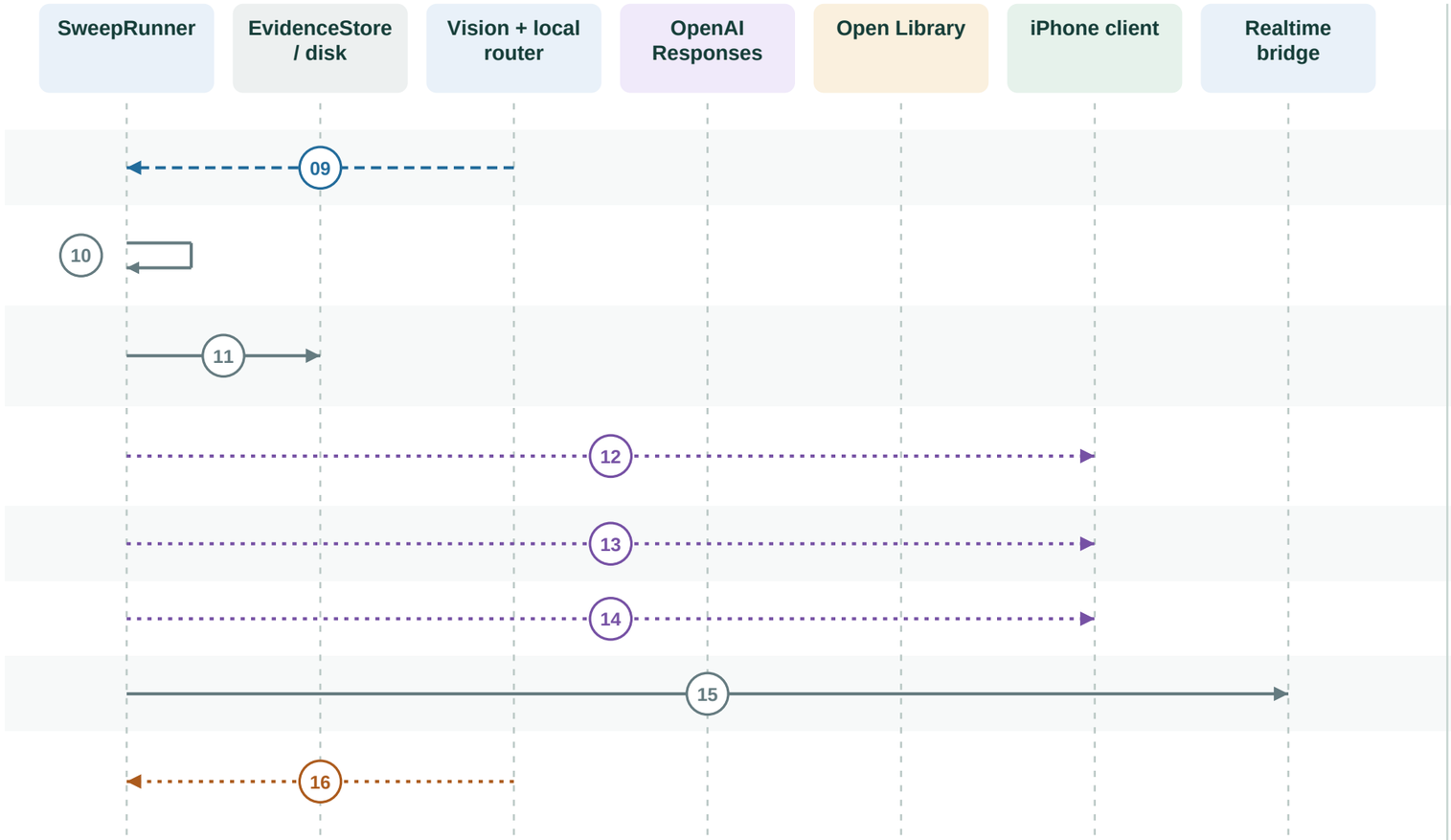
08 Completed parsed output + usage
Detections, polygons, visible text, quality issues, guidance; no price/scale authority.

BOUNDARY Current phone produces no local semantic book clues; Open Library is a compatibility branch, not the normal current path. Quality/budget/cooldown can defer cloud. Model labels do not set dimensions or prices.

SOURCE claim_agent/pipeline.py:237 | claim_agent/local_evidence.py:67 | claim_agent/local_evidence.py:161 | claim_agent/openai_provider.py:24 | claim_agent/pipeline.py:286

F05 Recognize a frame and publish inventory - part 2/2

Frame workers run concurrently; publication order can differ from capture order.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

09 Validated scan + routing + quality
Restore raw-image coordinates; reject malformed individual polygons.

10 Under lock: geometry, association and apply_scan
Measure sensor-supported faces; match physical records; gate book identity on OCR.

11 Save observation/overlays; revise state
Increment revision, persist immutable refs and atomic state; queue eligible prices.

12 inventory {packet} via API WebSocket
Full saved snapshot, including usage/cost estimate and unresolved review entries.

13 frame_overlays then recognition_status
Overlay geometry; status ready/pending, routing, cost and provisional objects.

14 Optional guidance via API WebSocket
User-facing capture guidance, separate from structured voice context.

15 offer_context: saved recognition facts
Grounded text context for Realtime (F03); no routine JPEG copied to voice.

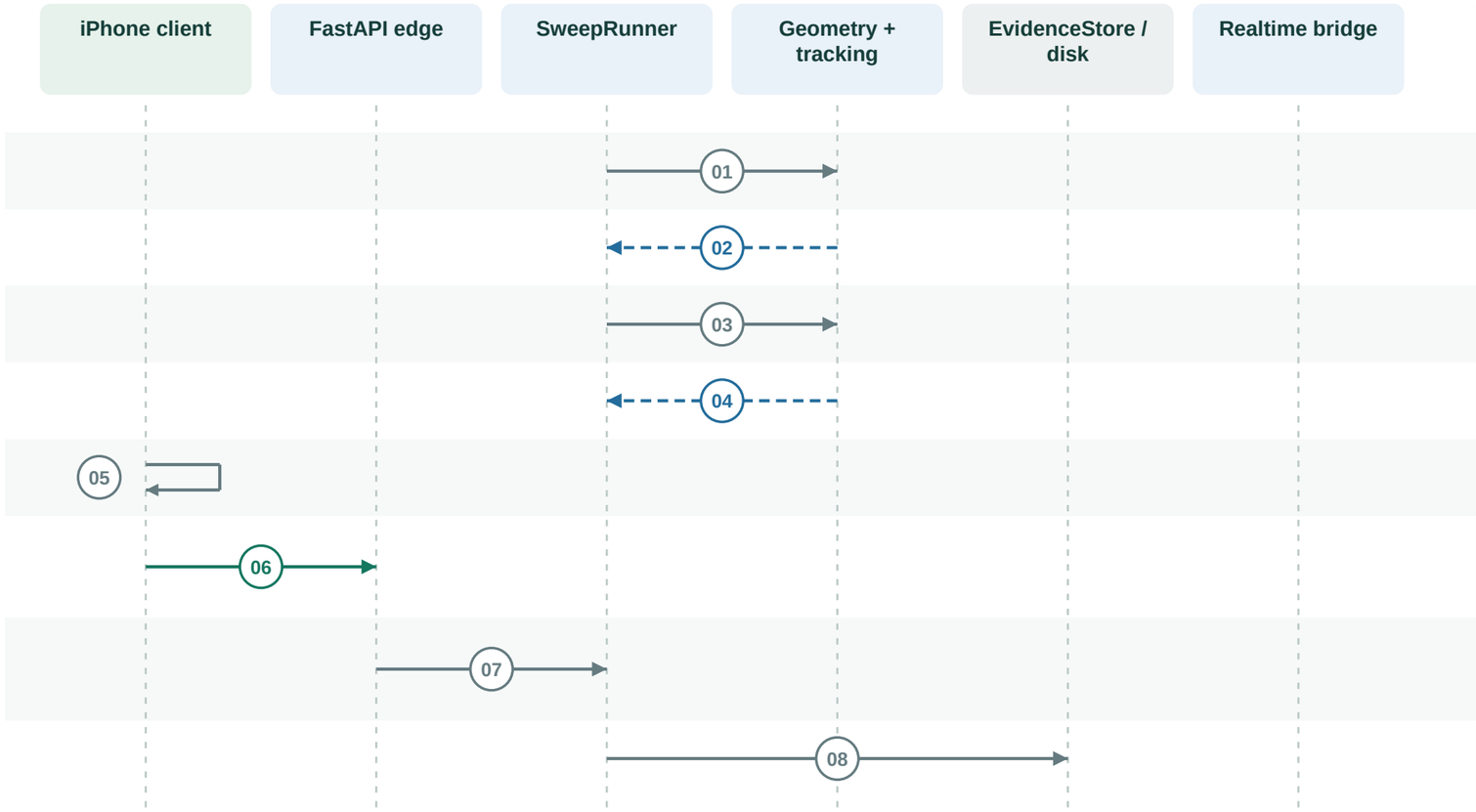
16 ALT deferred / provider failure / worker failure
Retain supported evidence; unknown usage is null; emit pending or unavailable status.

BOUNDARY Current phone produces no local semantic book clues; Open Library is a compatibility branch, not the normal current path. Quality/budget/cooldown can defer cloud. Model labels do not set dimensions or prices.

SOURCE claim_agent/pipeline.py:237 | claim_agent/local_evidence.py:67 | claim_agent/local_evidence.py:161 | claim_agent/openai_provider.py:24 | claim_agent/pipeline.py:286

F06 Measure objects and update RoomPlan geometry - part 1/2

Object face measurements and room/furniture geometry use calibrated sensor evidence, not guessed model scale.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 For each scan polygon: measure detection
Use original calibration and confident aligned depth; fit local plane/rays.

02 Dimensions / world support or unknown + reasons
Book spine edges or visible object face; no invented full 3-D box.

03 Associate physical track and furniture anchor
Same session, one-to-one assignment, compatible category/depth/geometry.

04 Existing physical ID or new/reviewable record
Text equality alone never merges physical objects; ambiguity stays explicit.

05 RoomPlan live update on shared AR session
Throttle >=3 s; retain latest snapshot; only one geometry request in flight.

06 POST /api/sweeps/{id}/geometry
After accepted frame: session ID, walls/openings/objects; live floors:[], capture_complete:false.

07 Validate session and geometry write state
Camera mode, absent first frame or mismatched session reject; same bytes are idempotent.

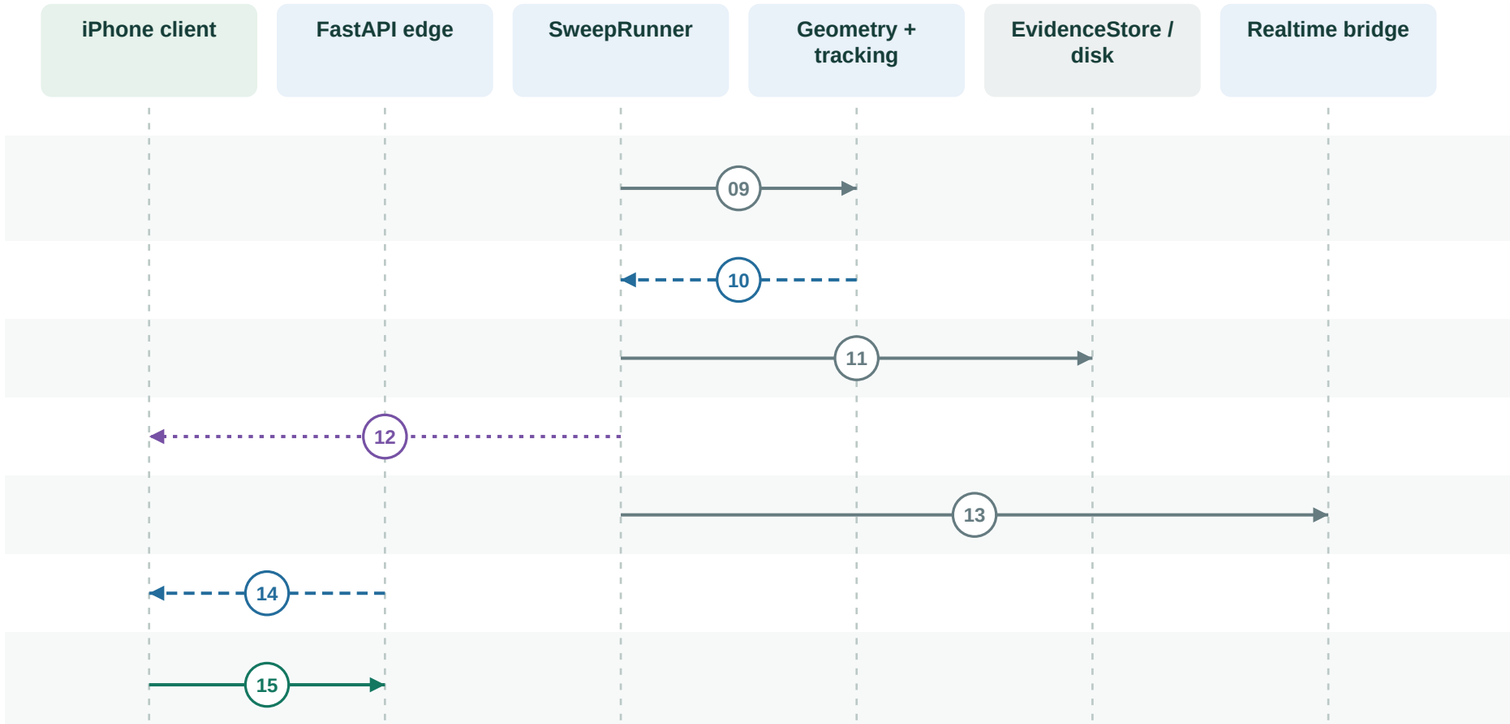
08 Save immutable geometry/room_<id>.json
Preserve the input geometry before reconciling measurements.

BOUNDARY Live RoomPlan intentionally omits floor polygons. Final RoomBuilder geometry is needed for floor area. Camera-only frames keep unsupported measurements unknown. Accurate sensor-derived numbers still need tape/ruler validation.

SOURCE claim_agent/pipeline.py:369 | claim_agent/geometry.py:30 | ios/LibraryClaim/MeasurementModels.swift:4 | ios/LibraryClaim/RoomGeometry.swift:5 | claim_agent/pipeline.py:711

F06 Measure objects and update RoomPlan geometry - part 2/2

Object face measurements and room/furniture geometry use calibrated sensor evidence, not guessed model scale.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

09 measure_room + reconcile furniture
Floor/wall polygons, openings, wall height, supported RoomPlan boxes, one-to-one matches.

10 Room facts / furniture dimensions / review issues
Curved or incomplete surfaces and ambiguous links preserve unknowns.

11 Update state; refresh affected price queries
Clear obsolete dimensions on revision; stale quotes cannot apply.

12 inventory {packet} via API WebSocket
Event update can arrive before the geometry HTTP response.

13 offer_context: measurement facts
Optional grounded voice cue; this is an internal bridge call, not a phone event.

14 200 {saved:true}
422 unsupported/session-invalid; 409 changed geometry after capture ended.

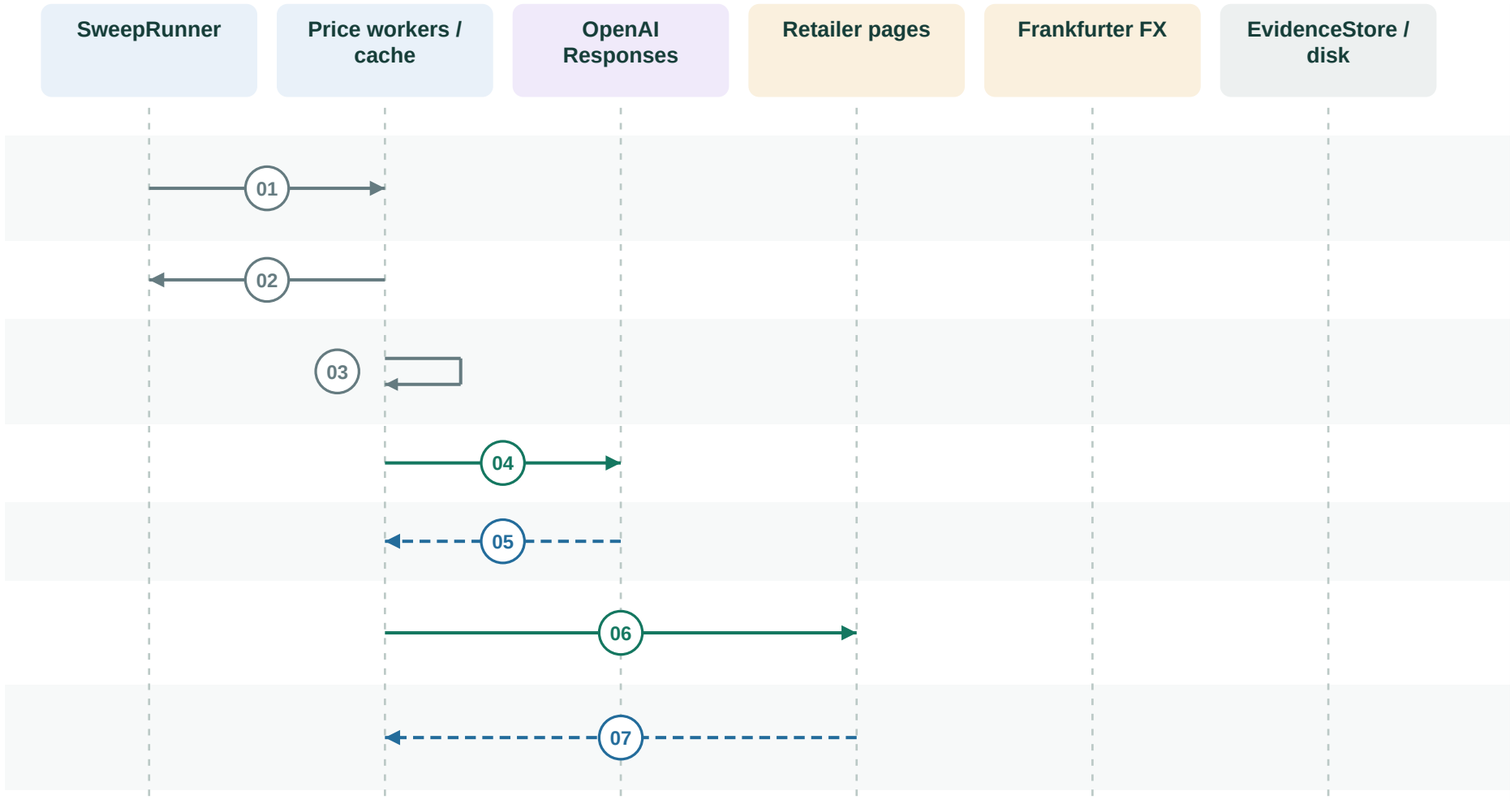
15 At end, if available: POST final geometry
LiDAR RoomBuilder output before /finish; failures become capture issues and geometry may be skipped.

BOUNDARY Live RoomPlan intentionally omits floor polygons. Final RoomBuilder geometry is needed for floor area. Camera-only frames keep unsupported measurements unknown. Accurate sensor-derived numbers still need tape/ruler validation.

SOURCE claim_agent/pipeline.py:369 | claim_agent/geometry.py:30 | ios/LibraryClaim/MeasurementModels.swift:4 | ios/LibraryClaim/RoomGeometry.swift:5 | claim_agent/pipeline.py:711

F07 Search, verify and apply a price - part 1/2

Pricing is a separate in-process queue. Search returns URLs; fetched source offers establish price evidence.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 Enqueue eligible record + revision/fingerprint
Books: new and used separately; items: new only. No repeat of unchanged attempted job.

02 Set queued -> searching; save/publish inventory
Check eligibility, current revision, paid-pricing setting and provider latch.

03 Check quote/search cache and shared in-flight task
Quote cache skips network; search cache skips model search only. Per-runner caches; raw current query validated.

04 POST /v1/responses with web_search
gpt-5.4-mini, country/currency/condition query, required search, one tool call.

05 Completed search citations/sources + usage
Up to eight candidate URLs; generated price prose is not accepted.

06 GET discovered public product URLs
Fetch candidates concurrently; revalidate redirects/addresses; timeout/size limits.

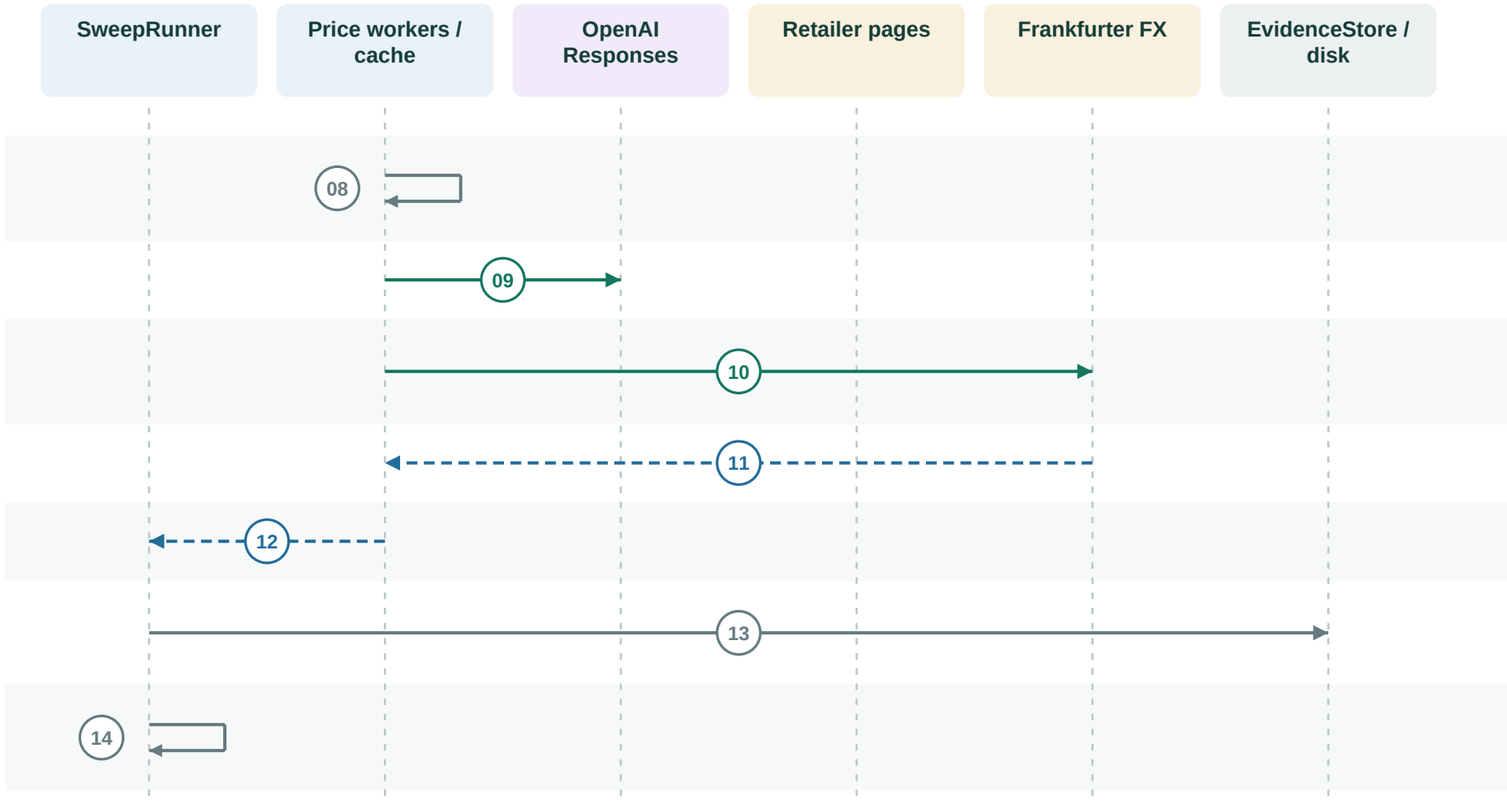
07 HTML/text containing JSON-LD Product/Book Offer
No fixed retailer pricing API; blocked/unstructured sources may give no usable offer.

BOUNDARY Quote-cache hit skips network calls; search-cache hit skips only model search and still fetches/validates retailers. Caches/in-flight sharing are per runner. A quote has a 45 s deadline after semaphore entry. Unknown offers stay outside totals. API USD usage costs are separate from local-currency claim values.

SOURCE claim_agent/pipeline.py:551 | claim_agent/pipeline.py:620 | claim_agent/pricing.py:350 | claim_agent/openai_provider.py:71 | claim_agent/providers.py:18 | claim_agent/pricing.py:466

F07 Search, verify and apply a price - part 2/2

Pricing is a separate in-process queue. Search returns URLs; fetched source offers establish price evidence.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

08 Validate and rank individual offers
Identity, condition, stock, market/currency, material/dimensions and retrieval date.

09 OPTIONAL foreign-market fallback search
If local sources fail, search UK/GBP or India/INR; repeat retrieval/validation.

10 OPTIONAL GET /v2/rate/{base}/{quote}
Foreign conversion and non-INR appraisal-threshold check; pair cache may satisfy.

11 JSON {base, quote, date, rate}
Validate dated positive rate and pair; preserve original amount/currency.

12 QuoteResult with Price and source metadata
Or amount:null + explicit unavailable/timeout/access reason.

13 Revalidate revision; persist offer/search/FX refs
Reject stale results; update pricing status; no automatic retry after uncertain spend.

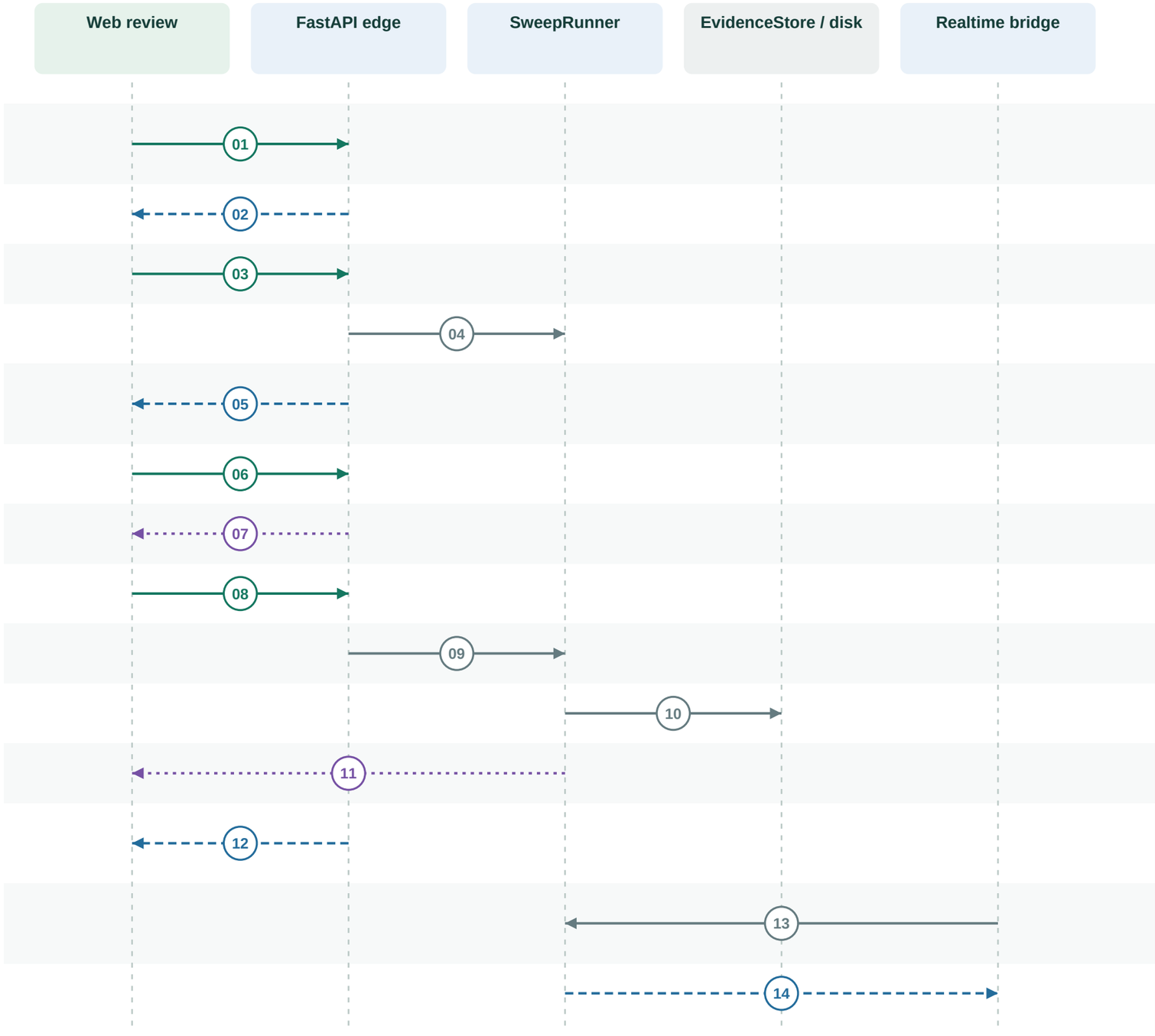
14 apply_quote -> save_publish -> optional bundle refresh
Later inventory event and valuation_updated voice context announce current state.

BOUNDARY Quote-cache hit skips network calls; search-cache hit skips only model search and still fetches/validates retailers. Caches/in-flight sharing are per runner. A quote has a 45 s deadline after semaphore entry. Unknown offers stay outside totals. API USD usage costs are separate from local-currency claim values.

SOURCE claim_agent/pipeline.py:551 | claim_agent/pipeline.py:620 | claim_agent/pricing.py:350 | claim_agent/openai_provider.py:71 | claim_agent/providers.py:18 | claim_agent/pricing.py:466

F08 Read inventory and apply corrections

Browser review pairs with a cookie; native HTTP uses a bearer token. Both can subscribe to the same event route.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 POST /api/pair {token}
Optional browser bootstrap; token supplied from review link fragment, then cleared.

02 200 {paired:true} + claim_pair cookie
HttpOnly, SameSite=Strict; Secure only over HTTPS. Bad token: 401.

03 GET /api/sweeps; GET /api/sweeps/{id}
List latest 20 saved entries; select one for a current snapshot.

04 runner_for(id): lazy load if needed
After restart mark interrupted capture partial; resume eligible pricing only.

05 200 list / 200 ClaimPacket
Packet includes sweep, room, books/items, totals, review_queue, revision, metrics.

06 WS /api/sweeps/{id}/live
Browser can stay event-only; it does not send start_voice.

07 Initial inventory, then updated packets/events
Initial snapshot and later inventory use {type:"inventory",packet}.

08 POST /api/sweeps/{id}/corrections
JSON: action, ref_id, value. Use an exact saved inventory/shelf ID.

09 correct(): resolve alias, validate and revise
Invalidate affected price jobs/values; append owner claim/exclusion audit.

10 Persist revised state and refresh finished bundle
Print confirmation may enable pricing; unsupported claims require appraisal.

11 inventory {packet} via API
State publication occurs before the correction HTTP response.

12 200 updated ClaimPacket
Invalid reference/action: 422; identical correction is not automatically deduplicated.

13 ALTERNATIVE voice: correct_inventory tool
Validated provider function event invokes the same method, not another HTTP service.

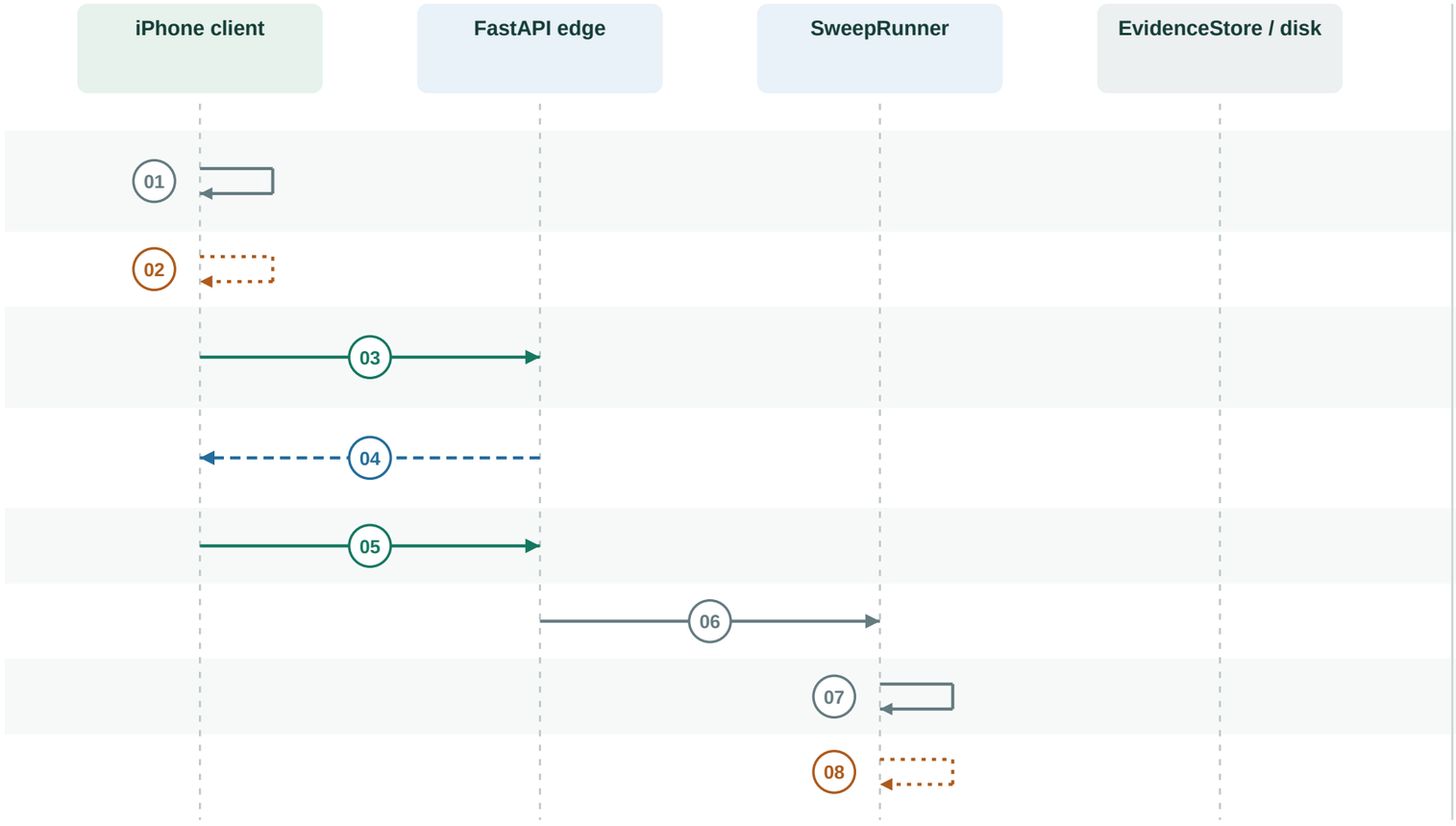
14 Internal snapshot/result -> function_call_output
Tool result is sent back to Realtime by call_id; see F02/F03.

BOUNDARY There is no SSE endpoint or durable event cursor. Listener overflow drops oldest events; GET or a new WebSocket snapshot recovers current state. Lazy-loading a saved sweep may resume eligible pricing, but not frame jobs.

SOURCE claim_agent/static/app.js:28 | claim_agent/server.py:157 | claim_agent/server.py:193 | claim_agent/server.py:378 | claim_agent/pipeline.py:732 | claim_agent/live.py:65

F09 Finish, generate and download claim evidence - part 1/2

The native client flushes saved evidence and final geometry before a held /finish request.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 Stop sampling; LiDAR RoomBuilder if supported
Await encoder; save final geometry if available; drain pending uploads for up to 60 s.

02 ALT pending evidence remains: Awaiting uploads
Expose Retry finish; do not call server finish while pending files remain.

03 If available: POST /geometry after live drain
Send final LiDAR geometry when present; camera or failed export skips it with capture issues.

04 200 {saved:true}
Require success when geometry exists; otherwise finalize with recorded geometry-failure issues.

05 POST /api/sweeps/{id}/finish
duration_s and capture_issues; native request timeout 300 s.

06 finish_lock; transition to processing
Already ready/partial returns snapshot. Otherwise save/publish processing state.

07 Await frame queue, then price queue
Workers run concurrently; bounded overall finish_timeout, source default 240 s.

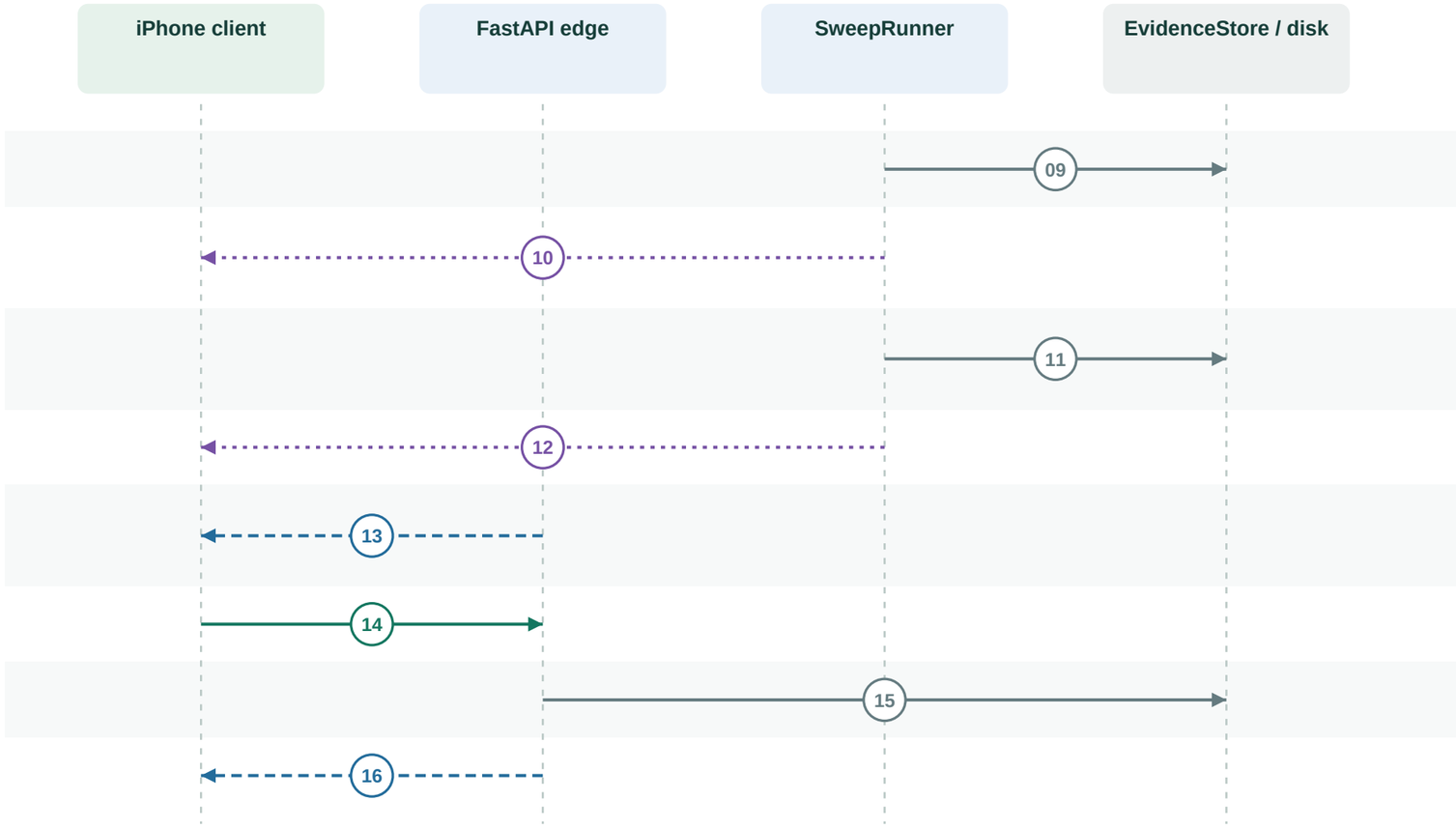
08 ALT deadline: cancel remaining workers
Mark unfinished pricing unknown/reviewable; continue to partial packet.

BOUNDARY Finish returns 200 only after drain/deadline handling; it is not a 202 job receipt. Partial can still be a successful HTTP outcome. Exports may also be requested before finish. Comparison is not automatically inside packet.zip.

SOURCE ios/LibraryClaim/CaptureController.swift:487 | ios/LibraryClaim/CaptureController.swift:519 | claim_agent/pipeline.py:776 | claim_agent/packet.py:150 | claim_agent/server.py:397

F09 Finish, generate and download claim evidence - part 2/2

The native client flushes saved evidence and final geometry before a held /finish request.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

09 Compute Decimal totals; save final state
Preserve ready/partial state, review queue and immutable evidence refs.

10 inventory {packet} via API
Final snapshot is published before bundle generation; it does not prove the ZIP exists.

11 Build bundle; update timing; save/rewrite bundle
Generate JSON, HTML and ZIP; then offer capture-ended context to internal voice bridge.

12 packet_ready {packet} via API
Emitted after bundle generation; final HTTP response follows.

13 200 final/partial ClaimPacket
Held HTTP response completes; this does not prove all acceptance bars passed.

14 GET /api/sweeps/{id}/packet.zip
Optional alternatives: claim_packet.json, report.html, evidence path.

15 Rebuild bundle under lock; verify evidence hashes
ZIP includes packet/report/manifest and registered evidence, not raw state.json.

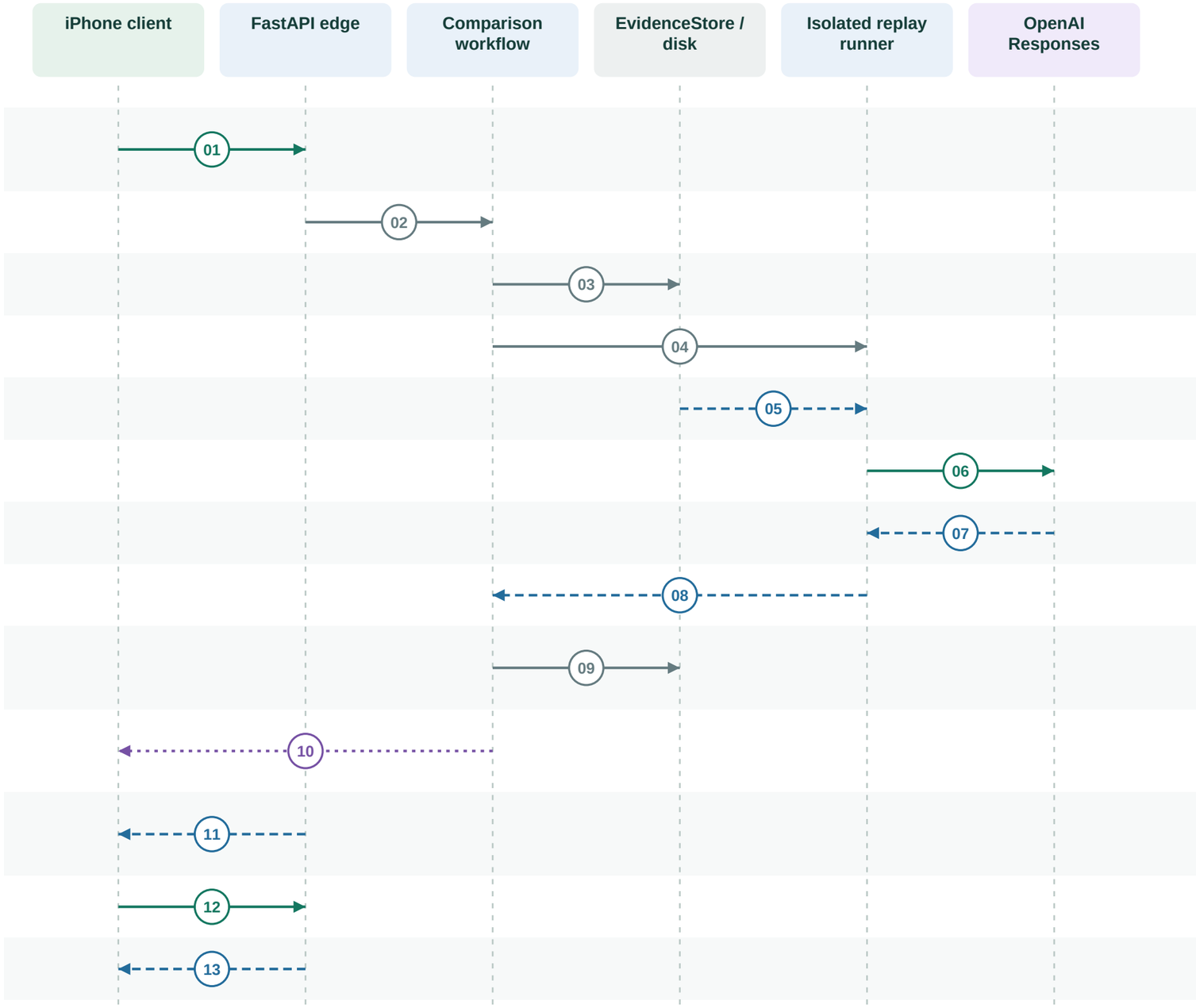
16 200 application/zip; filename claim_{id}.zip
Native saves local download; optional comparison starts afterwards (F10).

BOUNDARY Finish returns 200 only after drain/deadline handling; it is not a 202 job receipt. Partial can still be a successful HTTP outcome. Exports may also be requested before finish. Comparison is not automatically inside packet.zip.

SOURCE ios/LibraryClaim/CaptureController.swift:487 | ios/LibraryClaim/CaptureController.swift:519 | claim_agent/pipeline.py:776 | claim_agent/packet.py:150 | claim_agent/server.py:397

F10 Compare camera-only with LiDAR

Optional same-frame depth ablation: one held POST, persisted status reads, and a separate replay runner.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 POST /api/sweeps/{id}/camera-comparison
JSON {}; native waits up to 300 s. Must follow eligible ready/partial LiDAR sweep.

02 Acquire filesystem lock; inspect previous report
Existing completed/failed report returns without another replay.

03 Freeze LiDAR packet; persist running status
Validate model/provider continuity, evidence and total active capacity.

04 Create fresh camera runner / provider / state
Same locale/model and original RGB+calibration/pose; zero depth dimensions.

05 Original hash-verified JPEGs and calibration
Do not inherit source tracks, physical IDs, corrections, geometry or prices.

06 Gated vision/search requests (F05/F07)
Independent call budget and costs; process each frame, then finish replay.

07 Independent observations / sources / usage
Unknown/error outcomes remain distinct from zero detected objects.

08 Camera packet and processing evidence
Counts, measurement coverage, review reasons, timing and usage.

09 Persist camera packet and comparison report
status complete or failed; analysis_status separately completed/partial/failed/interrupted.

10 comparison_ready {report} via API
Completed/handled failure emits event. Cancellation may persist failure without event; recover through GET.

11 200 saved report after processing
No 202 response. 409 already_running/source_unavailable/capacity before replay.

12 OPTIONAL GET /api/sweeps/{id}/camera-comparison
Read persisted status; 404 if absent. No replay starts from this GET.

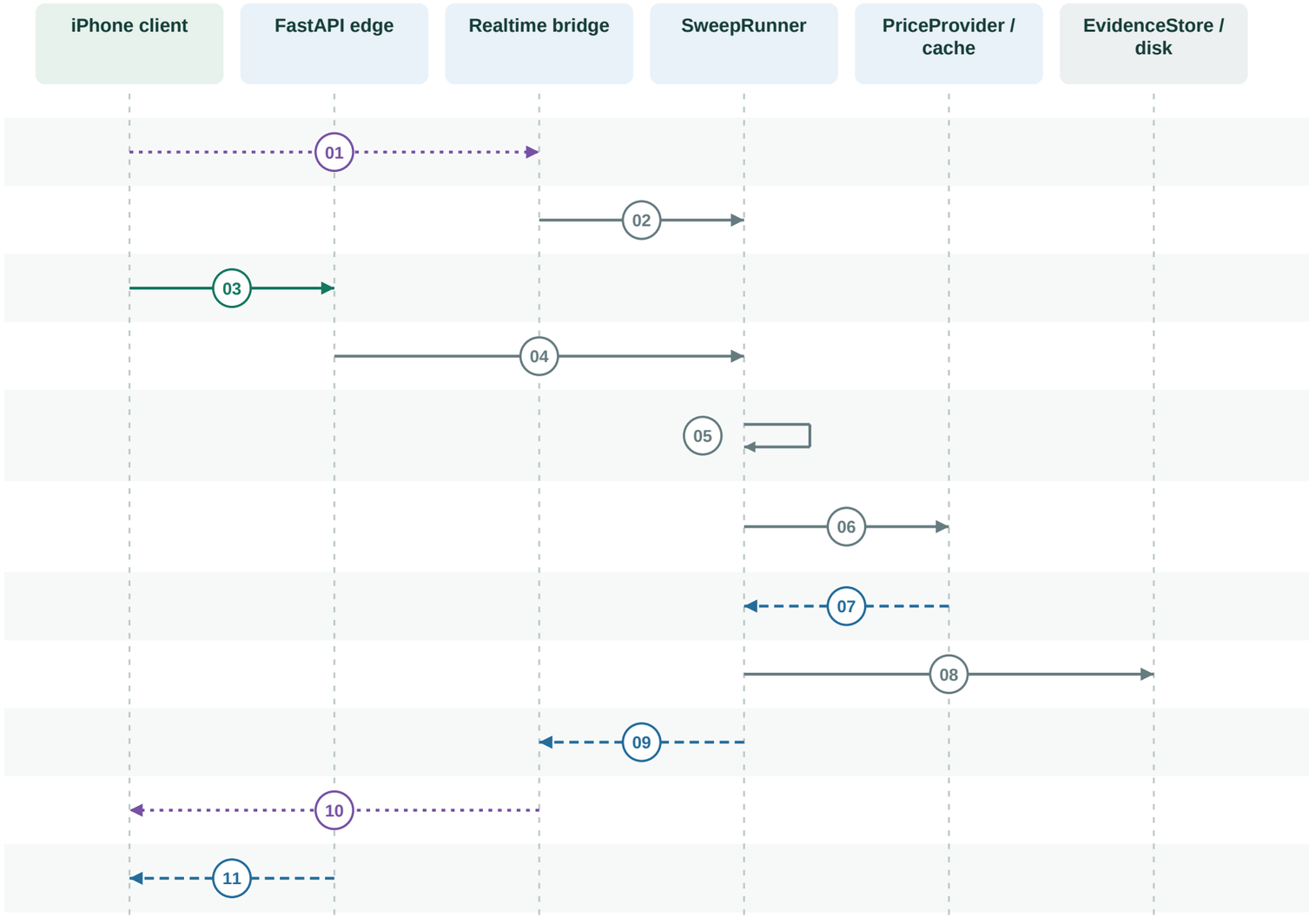
13 200 persisted running/complete/failed report
Already-running native recovery does one GET; no automatic polling loop.

BOUNDARY This reuses JPEGs and AR poses, not source depth/geometry/identity/prices. Poses may have benefited from LiDAR. A complete workflow can contain partial/failed analysis; it is not an accuracy or sensor-speed benchmark.

SOURCE claim_agent/comparison.py:107 | claim_agent/comparison.py:115 | claim_agent/comparison.py:152 | ios/LibraryClaim/BackendClient.swift:119

F11 Reprice the same books in a second market

Two entry points share one method: voice emits a result event; direct HTTP returns the result without that event.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 **PATH A: spoken request through Realtime (F03)**
Validated compare_market function arguments: country, currency.

02 **compare_market -> second_locale(locale)**
Validate supported voice country/currency pair; no separate HTTP round trip.

03 **PATH B: POST /api/sweeps/{id}/second-locale**
JSON: country, currency; request stays open until results are ready.

04 **second_locale(locale)**
This is an alternative entry to step 2, not a second call after voice.

05 **Copy first ten identified, nonexcluded books**
Freeze same physical IDs and frame refs; report required=10 and actual coverage.

06 **For each selected book: quote new, then used**
Direct sequential quote calls with F07 validation; bypasses pricing-job queue and its resume/dedup rules.

07 **Prices with sources or explicit unknowns**
Do not overwrite primary-market prices or generate missing identities.

08 **Save source evidence and second-locale observation**
Persist state; no inventory revision/event or bundle regeneration here.

09 **PATH A result: same_physical_ids:true + books**
Bridge receives result after awaited pricing.

10 **PATH A only: second_locale {result}**
Also return function output to Realtime so spoken response can use result.

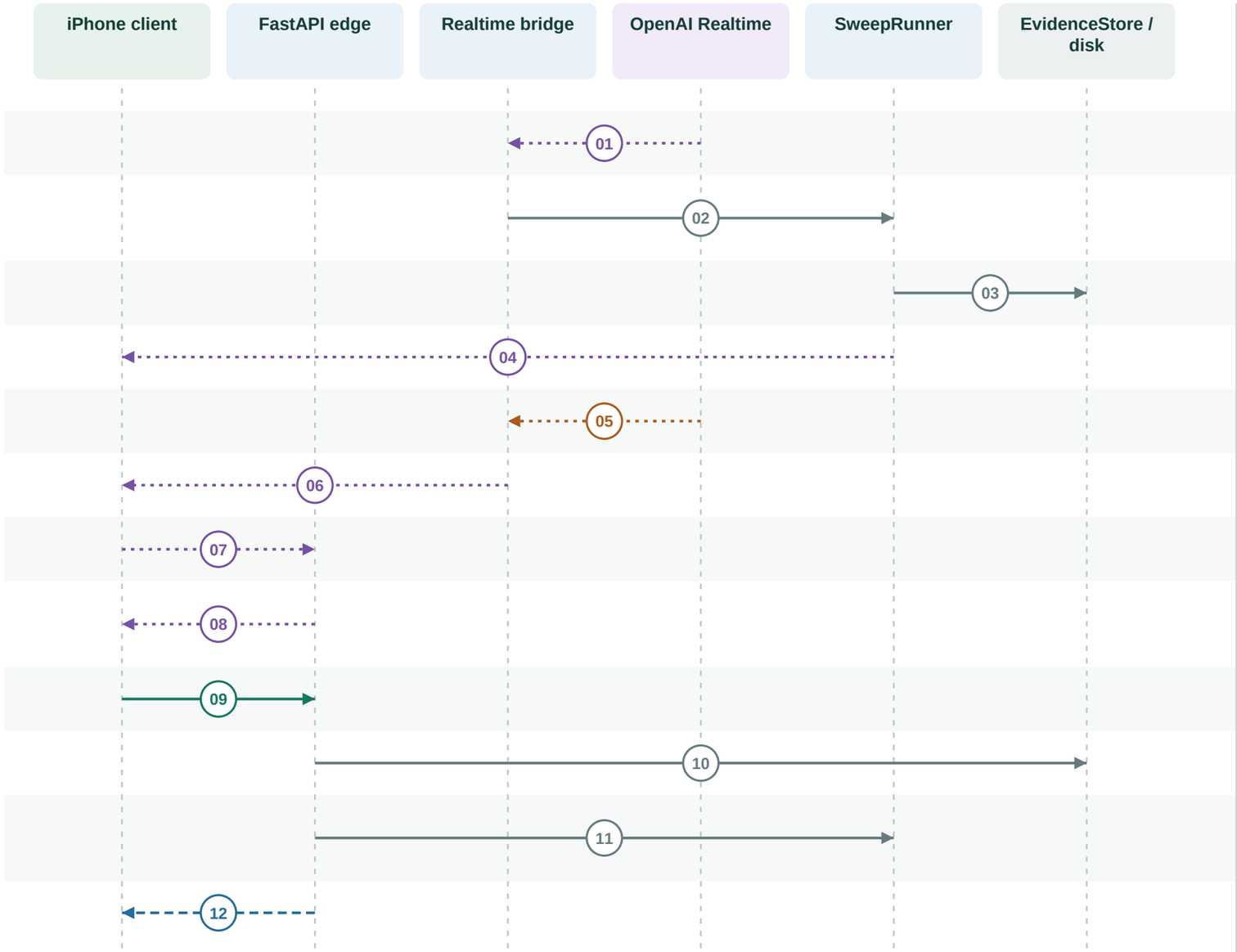
11 **PATH B only: 200 second-locale result**
No second_locale WebSocket event is emitted by this HTTP route.

BOUNDARY Copies the first ten eligible identified books, or fewer with coverage. Original IDs/frame refs stay the same; primary market prices are unchanged. Direct HTTP validates Locale syntax; voice also validates supported market pairing.

SOURCE claim_agent/server.py:394 | claim_agent/live.py:73 | claim_agent/pipeline.py:858 | ios/LibraryClaim/CaptureController.swift:698

F12 Reconnect, recover and account for usage

Recovery is bounded and explicit. Files are durable; provider sessions and most processing queues are not.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 **response.done with response.usage; transcription usage**
Record actual provider usage once per model/response ID.

02 **persist_usage -> save_publish**
Sweep path persists metrics; onboarding usage stays in memory until copied at handoff. Unknown usage:null.

03 **Atomic state + cost ledger snapshot**
Compute known estimated subtotal and uncosted count; unknown is not zero.

04 **inventory {packet} via API**
Metrics are nested inside packet; no dedicated usage event.

05 **ALT provider access/credit or fatal voice error**
Selected stale control errors recover; other failures terminate voice bridge.

06 **voice_error; review reason saved**
Request explicit reconnect/funding action; retain saved camera evidence.

07 **User reconnect: close old WS, open new WS**
Repeat health/readiness negotiation in F01; no background paid reconnect loop.

08 **Current snapshot and newly negotiated voice**
Unexpired in-memory setup can survive reconnect, not server restart. Prior provider audio buffers do not.

09 **After server restart: GET saved sweep**
runner_for loads state; saved capturing/processing changes to partial.

10 **Load persisted state and evidence**
No durable frame-task broker; missing capture continuity is reviewable.

11 **resume_pricing with identity/spend guards**
Queued eligible jobs may resume; searching becomes interrupted without replay.

12 **200 current ClaimPacket with review/unknowns**
Later price publications use the same event route; no claimed full recovery.

BOUNDARY No blanket automatic paid retry. Persisted searching price jobs become interrupted/unknown-cost; eligible queued prices may resume. Server restart does not replay saved frame jobs. Costs can remain partial even when a packet exists.

SOURCE claim_agent/server.py:157 | claim_agent/pipeline.py:61 | claim_agent/openai_live.py:238 | claim_agent/openai_live.py:694 | claim_agent/costing.py:57 | ios/LibraryClaim/CaptureController.swift:742

Endpoint and contract index

Active routes and optional workflows; legacy voice HTTP routes remain implemented but are not used by the active native Realtime flow.

Method	Route	Result / transport	Purpose / flow
GET	/api/health	200 configuration JSON	F01; unauthenticated; no paid readiness probe
POST	/api/pair	200 {paired:true}; cookie	F08 browser-only pairing; 401 for bad token
POST	/api/onboarding	200 {id,expires_in_s}; 409 capacity	F01; JSON device/capture_supported/camera_supported
WS	/api/onboarding/{id}/live	Accepted + async setup/audio events	F01-F03; 4401/4404 are pre-accept rejection branches
POST	/api/sweeps	200 ClaimPacket; 409 invalid handoff	F02; native {device,onboarding_id}; retry reuses saved handoff
GET	/api/sweeps	200 summary array, latest 20	F08; reads state files, does not instantiate runners
GET	/api/sweeps/{id}	200 current ClaimPacket	F08/F12; lazy load may resume eligible pricing
WS	/api/sweeps/{id}/live	Initial inventory; async events	F03/F05/F08; no SSE; voice optional on this socket
POST	/api/sweeps/{id}/frames	200 receipt / 429 backpressure	F04; metadata + JPEG/depth/confidence base64; result later via WS
POST	/api/sweeps/{id}/geometry	200 {saved:true}	F06; same-session geometry; invalid/session errors 422
POST	/api/sweeps/{id}/corrections	200 updated ClaimPacket	F08; {action,ref_id,value}; inventory published before response
POST	/api/sweeps/{id}/finish	200 ready/partial ClaimPacket	F09; held request, not 202; duration_s/capture_issues
POST	/api/sweeps/{id}/second-locale	200 comparison result	F11; held request; does not itself emit second_locale event
GET	/api/sweeps/{id}/camera-comparison	200 persisted report; 404 absent	F10; reads running/complete/failed; does not start replay

Common errors: 401 pairing, 403 Origin, 413 body size, 422 validation; operation-specific alternatives appear in each flow.

Endpoint and contract index

Active routes and optional workflows; legacy voice HTTP routes remain implemented but are not used by the active native Realtime flow.

Method	Route	Result / transport	Purpose / flow
POST	/api/sweeps/{id}/camera-comparison	200 report; 409 conflict/capacity	F10; held request; reuses saved result without extra replay
GET	/api/sweeps/{id}/packet.zip	200 application/zip attachment	F09; rebuild current bundle; not gated on finished state
GET	/api/sweeps/{id}/claim_packet.json	200 current ClaimPacket JSON	F09; computed snapshot rather than reading stored file
GET	/api/sweeps/{id}/report.html	200 text/html	F09; regenerate bundle then serve readable report
GET	/api/sweeps/{id}/{kind}/{name}	200 JPEG or octet-stream	F09; allowlisted path and saved SHA-256 validation
GET/POST	/api/{onboarding|sweeps}/{id}/local-voice	Legacy context / tool contract	Inactive native path; do not confuse with OpenAI Realtime
POST	/api/{onboarding|sweeps}/{id}/local-voice/assistant	Legacy text assistant contract	Inactive native path; historical provider adapters remain
POST	/api/{onboarding|sweeps}/{id}/local-voice/{transcribe|speech}	Legacy speech HTTP contracts	Inactive native path; no active sequential STT/TTS chain
POST	/api/sweeps/{id}/local-voice/guidance	Legacy spoken-guidance contract	Inactive native path; current bridge uses Realtime context
WSS	api.openai.com/v1/realtime?model=...	session/audio/transcript/tool/usage events	F01-F03; server credential; independent upstream socket
POST	api.openai.com/v1/responses	Structured scan or search URLs + usage	F05/F07; standard SDK base, configurable; store:false
GET	Discovered public retailer product URL	HTML/text with JSON-LD offers	F07; credential-free retrieval; redirect/address validation
GET	api.frankfurter.dev/v2/rate/{base}/{quote}	JSON pair/date/rate	F07; conversion and appraisal threshold; optional/cacheable
GET	openlibrary.org/search.json	Candidate bibliographic metadata	F05 compatibility branch; not normal current-phone path

Common errors: 401 pairing, 403 Origin, 413 body size, 422 validation; operation-specific alternatives appear in each flow.